

GSEB UNIT TEST

Standard 10th Science | Chapter 1: Chemical Reactions and Equations

STRICTLY TIME BOUND

Time: 2 Hours

Max. Marks: 50

IMPORTANT INSTRUCTIONS FOR HOME EXAM:

- Maintain complete silence and honesty as if you are in a board examination hall.
- Do not use AI tools, Google, textbooks, or any study material during the exam.
- All questions are compulsory and must be written in a clean, legible handwriting.
- Write balanced chemical equations with physical states wherever necessary.
- Draw neat diagrams with a pencil only where required.
- Check your answers thoroughly before submitting the paper to your supervisor/parent.

SECTION - A (Objective - 1 Mark Each)

1. What is the chemical formula of the brown gas evolved when Lead Nitrate is heated? [1]
2. Which substance is used for white-washing walls? [1]
3. State True/False: Electrolysis of water is a Physical Change. [1]
4. Define: Rancidity. [1]
5. Complete the reaction: $\text{CuO} + \text{H}_2 \rightarrow \underline{\hspace{2cm}} + \underline{\hspace{2cm}}$ [1]
6. What happens when CO_2 gas is passed through lime water for a short time? [1]
7. Which metal is used in the galvanization of iron? [1]
8. Identify the type: $\text{C} + \text{O}_2 \rightarrow \text{CO}_2 + \text{Heat}$ [1]
9. Why is Nitrogen gas filled in potato chips packets? [1]
10. Name the enzyme that helps in the process of digestion (an exothermic reaction). [1]

SECTION - B (Short Answer - 2 Marks Each)

11. Why should a magnesium ribbon be cleaned before burning in air? [2]
12. Balance: (i) $\text{HNO}_3 + \text{Ca(OH)}_2 \rightarrow \text{Ca(NO}_3)_2 + \text{H}_2\text{O}$ (ii) $\text{Al} + \text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + \text{H}_2$ [2]
13. Distinguish between Exothermic and Endothermic reactions with one example each. [2]
14. What is a decomposition reaction? Why is it called the opposite of combination? [2]
15. Explain the term 'Corrosion' with the help of the rusting of iron. [2]
16. What is observed when a solution of potassium iodide is added to lead nitrate? Write the equation. [2]

SECTION - C (Medium Answer - 3 Marks Each)

17. Explain Displacement Reaction by giving the example of Zn and Copper Sulphate solution. [3]
18. Write three methods to prevent the rancidity of food items containing fats and oils. [3]

19. Explain Redox reaction with a balanced equation. Identify the oxidizing and reducing agents. [3]
20. What is a double displacement reaction? Write a balanced equation for the reaction of Na_2SO_4 and BaCl_2 . [3]
21. Explain the decomposition of silver chloride in sunlight. Mention its use in photography. [3]
22. Write the chemical equations for: (i) Burning of Coal (ii) Formation of Water (iii) Respiration. [3]
- SECTION - D (Long Answer - 4/5 Marks Each)**
23. Describe the Electrolysis of Water with a labeled diagram and explain why the volume of one gas is double the other. [5]
24. Explain Thermal Decomposition in detail using the heating of Ferrous Sulphate crystals as an example. State the color change and the gases evolved. [5]

--- END OF PAPER | BE HONEST WITH YOURSELF ---